

ECOSORT AI

Smart Waste Segregation Assistant



1M1B AI for Sustainability Virtual Internship
In Collaboration with IBM SkillsBuild & AICTE

Submitted by:

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Primary SDG: SDG 12 – Responsible Consumption and Production
Secondary SDG: SDG 11 – Sustainable Cities and Communities

1. Abstract

EcoSort AI is a simple web-based prototype developed to help users correctly identify how to segregate everyday waste items. A user enters a short description of a waste item, and the system, using curated classification logic, returns the likely waste category, a disposal recommendation, the reason behind the classification, and a related sustainability tip. Hazardous and special-handling items are flagged clearly so that users are guided toward safe, authorized disposal methods rather than unsafe handling instructions.

The prototype was built using HTML, CSS, and JavaScript, without requiring a backend, database, paid APIs, or trained machine-learning models, making it realistic to develop and demonstrate within the timeline of the 1M1B AI for Sustainability Virtual Internship. It aligns with SDG 12 (Responsible Consumption and Production) as its primary goal and SDG 11 (Sustainable Cities and Communities) as a secondary goal, supporting these targets at a small, individual-awareness scale rather than claiming to solve them directly.

2. Introduction

2.1 Waste Segregation

Waste segregation is the practice of separating waste into categories such as wet/organic waste, dry waste, recyclable waste, e-waste, and hazardous waste, based on how each type should be processed or disposed of. Proper segregation at the source makes downstream waste management, recycling, and composting significantly more efficient.

2.2 Problems Caused by Incorrect Segregation

When waste is not segregated correctly, recyclable materials often end up in landfills, organic waste contaminates recyclables, and hazardous items such as batteries or e-waste may be mixed with general waste. This can reduce recycling efficiency, increase pollution risk, and create safety hazards for waste handlers.

2.3 Need for Simple Digital Guidance

Many people are unsure which category a given item belongs to, and official municipal guidelines are not always quick or convenient to consult at the exact moment of disposal. A simple, accessible digital tool that gives an instant recommendation can help bridge this gap in everyday awareness.

2.4 Why EcoSort AI Was Proposed

EcoSort AI was proposed as a lightweight, easy-to-build prototype that demonstrates how an AI-assisted decision-support workflow can guide users toward correct waste segregation. It was designed to be realistic and achievable within a short internship timeline, while still meaningfully addressing a genuine sustainability problem and following responsible AI practices.

3. Problem Statement

People often face difficulty in identifying the correct waste category for everyday items. Incorrect segregation can reduce recycling efficiency and create safety issues for special waste such as batteries and electronic devices.

“How might we use AI to help people correctly identify and segregate waste so that waste management becomes more sustainable?”

4. SDG Alignment

4.1 SDG 12 – Responsible Consumption and Production (Primary Goal)

EcoSort AI directly supports SDG 12 by encouraging responsible waste-management behavior, improving awareness of correct waste segregation, and promoting recycling and reduced landfill dependence at the individual and community level.

4.2 SDG 11 – Sustainable Cities and Communities (Secondary Connection)

Better waste segregation contributes to cleaner shared spaces, more efficient municipal waste collection, and safer handling conditions for waste workers, supporting the broader goal of sustainable, well-managed communities. EcoSort AI is a small-scale technology intervention that supports these SDGs; it does not claim to fully or directly solve them.

5. Target Users

EcoSort AI is designed for the following user groups:

Target User	Use
Students	Learn proper waste segregation
Households	Identify everyday household waste
Offices	Improve workplace segregation
Colleges	Promote sustainability awareness
Communities	Encourage responsible disposal

6. Objectives

- Develop a simple waste-segregation assistant.
- Identify common waste categories.
- Provide disposal recommendations.
- Explain why an item belongs to a category.
- Encourage sustainable waste-management practices.
- Demonstrate an AI-assisted sustainability concept.
- Support SDG 12 and SDG 11.

7. Design Thinking

Empathize ↓ Define ↓ Ideate ↓ Prototype ↓ Test & Refine

7.1 Empathize

Identified the difficulty users face when deciding how to dispose of common waste items, especially when an item does not clearly belong to one obvious category.

7.2 Define

Defined the problem as uncertainty in waste classification and disposal — users need a quick, reliable way to know how to handle a given waste item.

7.3 Ideate

Proposed a simple digital assistant that accepts a text description of a waste item and returns its category, disposal method, reasoning, and a sustainability tip, using an approach that is realistic to build without advanced infrastructure.

7.4 Prototype

Developed the EcoSort AI web application using HTML, CSS, and JavaScript, including the waste analyzer, classification logic, and supporting interface features described in Section 8.

7.5 Test & Refine

Tested representative waste items against the prototype and reviewed additional website features to confirm the interface behaved consistently and clearly communicated results to the user.

8. Prototype Features

The EcoSort AI prototype includes the following features:

- Waste analyzer
- Waste classification
- Recommended action
- Reason for classification
- Sustainability tip
- Hazardous-waste warning
- Example buttons
- Enter-key analysis
- Image upload interface
- Analysis history
- Eco points
- Eco level
- Dark mode

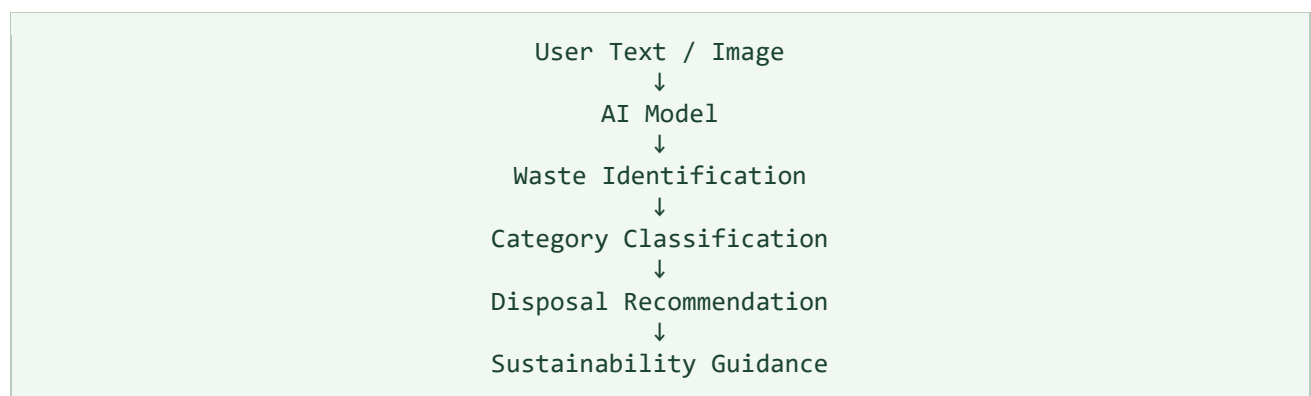
9. AI Methodology

The current EcoSort AI prototype demonstrates an AI-assisted decision-support workflow using curated classification logic for common waste descriptions. The system maps user inputs to predefined waste categories and provides corresponding recommendations, explanations, and sustainability tips.

The current image-upload functionality provides an image preview and is designed as a foundation for future multimodal AI integration. It does not currently perform image classification.

9.1 Future AI Pipeline

The diagram below shows how a future, more advanced version of EcoSort AI could process both text and image inputs using a trained AI model. This pipeline is proposed future scope and is not part of the current implementation.



10. System Architecture

The current prototype's architecture is illustrated below. All processing happens locally within the browser using JavaScript; no backend server or external database is used.



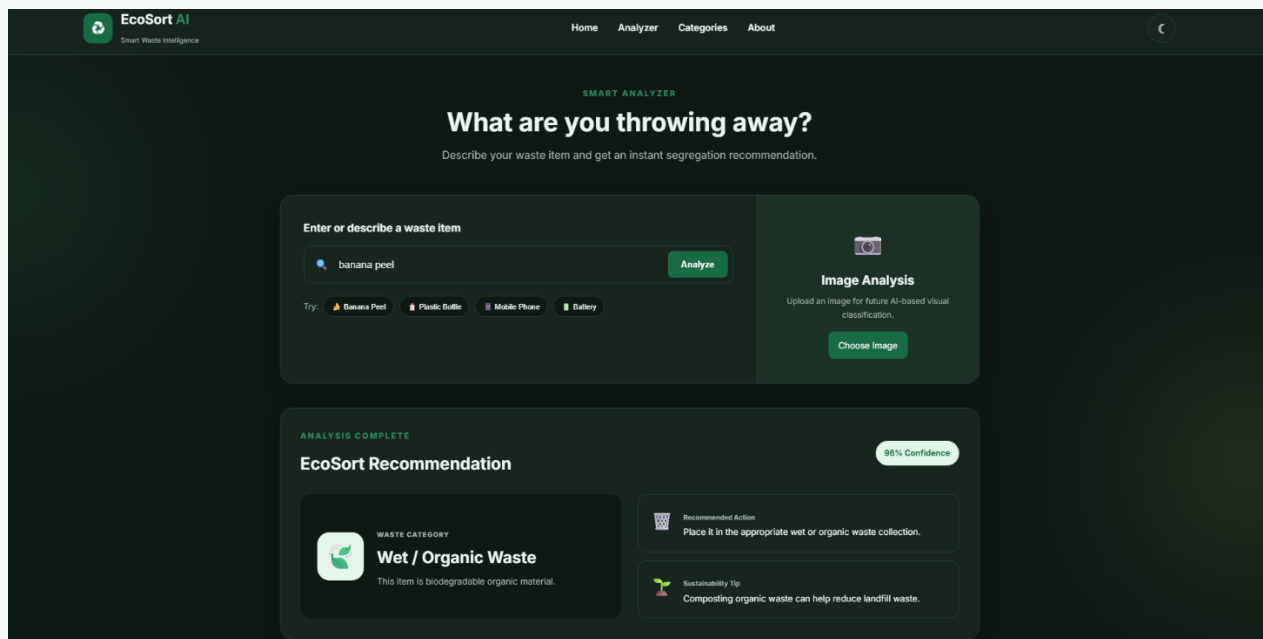
11. Technologies

Technology	Purpose
HTML5	Website structure
CSS3	Styling and interface
JavaScript	Application functionality
LocalStorage	Store analysis history
Web Browser	Run the application

12. Prototype Implementation

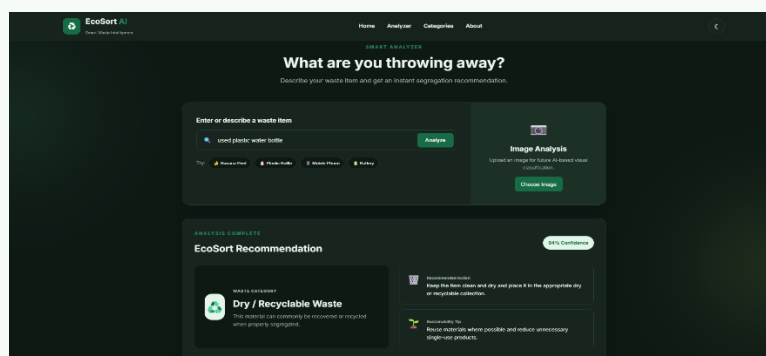
This section presents screenshots of the working EcoSort AI prototype. Screenshot placeholders are provided below — replace each placeholder box with the corresponding image before final submission.

12.1 Organic Waste Analysis



01-banana-peel.png

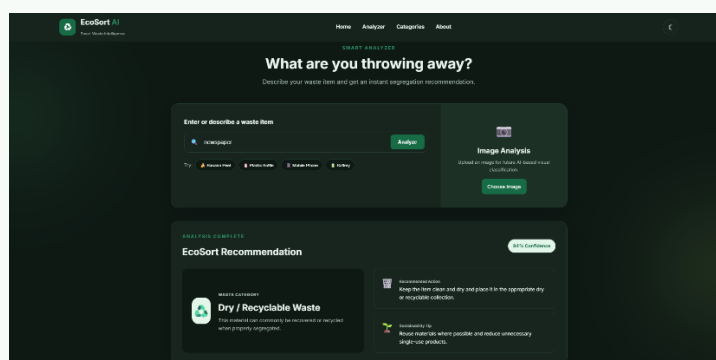
12.2 Recyclable Waste



02-plastic-bottle.png

Figure 2: Classification of used plastic water bottle as Dry / Recyclable Waste.

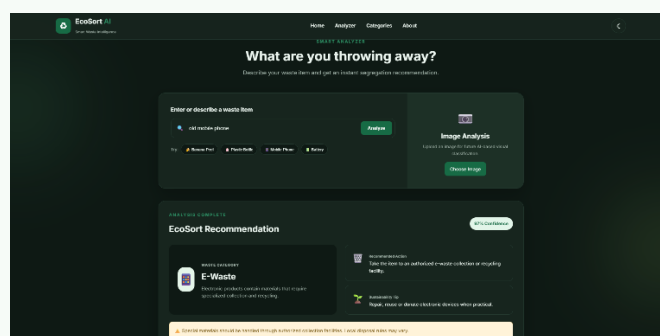
12.3 Newspaper Classification



03-newspaper.png

Figure 3: Newspaper classification.

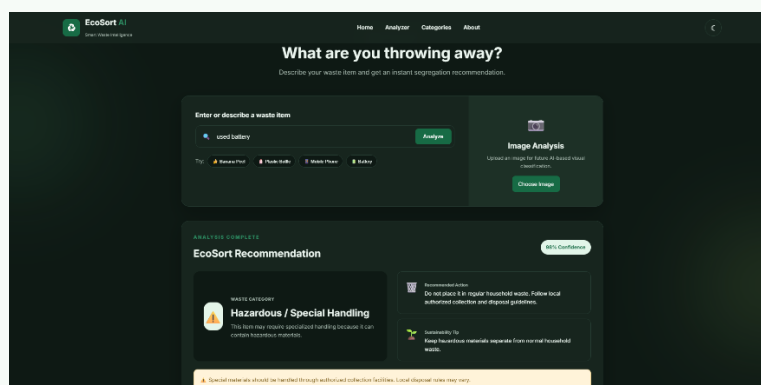
12.4 E-Waste Classification



04-mobile-phone.png

Figure 4: E-waste classification.

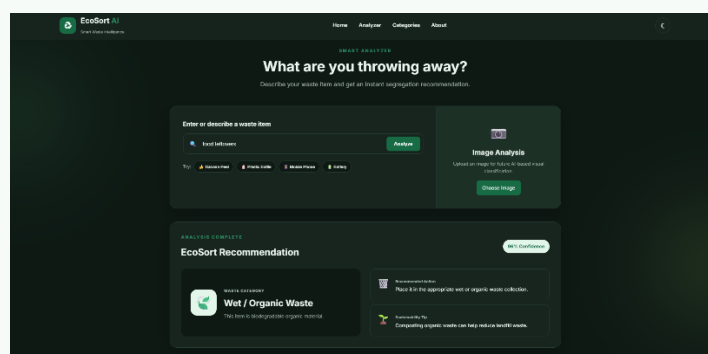
12.5 Hazardous Waste Classification



05-battery.png

Figure 5: Hazardous / special-handling classification.

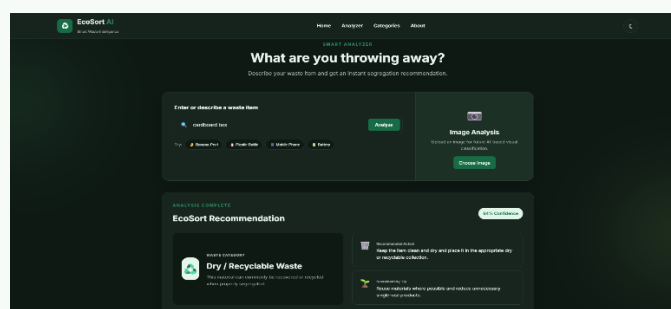
12.6 Food Waste Classification



06-food.png

Figure 6: Food waste classification.

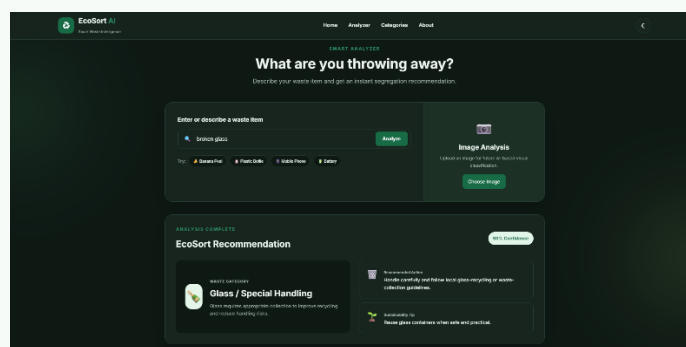
12.7 Cardboard Classification



07-cardboard.png

Figure 7: Cardboard classification.

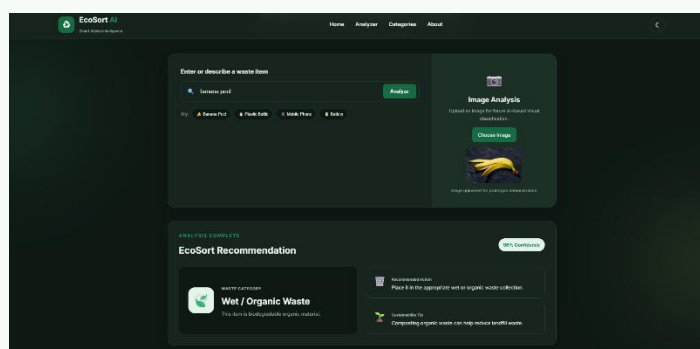
12.8 Glass Classification



08-glass.png

Figure 8: Glass / special-handling classification.

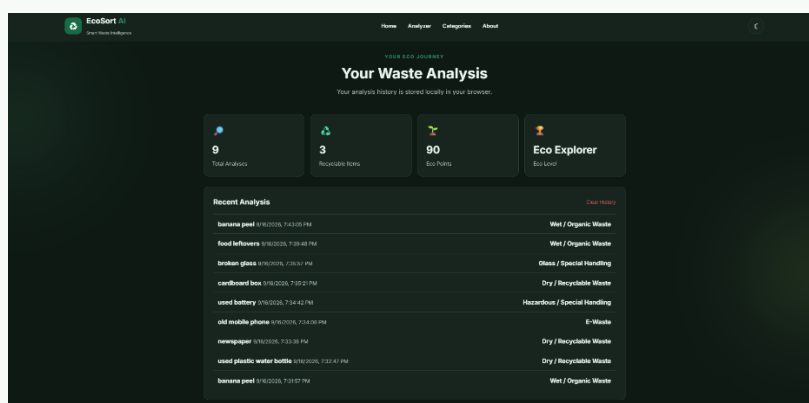
12.9 Image Upload Interface



09-image-upload.png

Figure 9: Image upload interface.

12.10 Dashboard and Analysis History



10-dashboard.png

Figure 10: Dashboard and analysis history.

12.11 Light Mode Interface

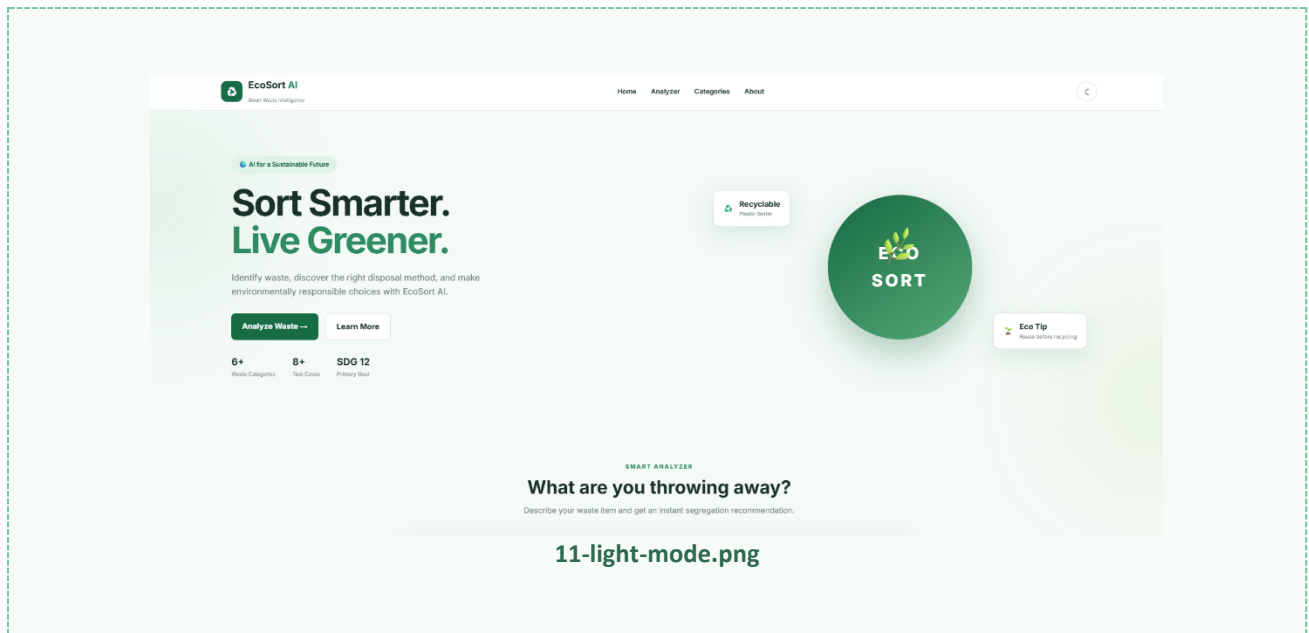


Figure 11: light-mode interface.

13. Testing

The prototype was tested against eight representative waste items. Each test compared the expected classification against the observed output of the EcoSort AI prototype.

Test Case	Input	Expected Result	Result
TC01	Banana peel	Wet / Organic Waste	Passed
TC02	Used plastic water bottle	Dry / Recyclable Waste	Passed
TC03	Newspaper	Dry / Recyclable Waste	Passed
TC04	Old mobile phone	E-Waste	Passed
TC05	Used battery	Hazardous / Special Handling	Passed
TC06	Food leftovers	Wet / Organic Waste	Passed
TC07	Cardboard box	Dry / Recyclable Waste	Passed
TC08	Broken glass	Glass / Special Handling	Passed

Prototype test-case result: 8 / 8 test cases passed on the demonstration examples above. This reflects consistency of the current rule-based/curated classification logic on these specific examples and is not equivalent to a real-world AI model accuracy measurement.

14. Responsible AI

Fairness

The classification logic applies consistently to similar waste descriptions, regardless of who is using the system.

Transparency

The current prototype clearly states that it uses curated classification logic rather than claiming a trained machine-learning model.

Privacy

The prototype minimizes data collection and uses browser-based local storage for history; no names, addresses, or other personal information is required.

Safety

Hazardous and special-handling items are flagged for additional care, and users are directed toward authorized disposal points rather than given unsafe handling instructions.

Human Oversight

Users should verify ambiguous items and follow local waste-management guidance. EcoSort AI is a decision-support tool, not a replacement for municipal waste-management authorities.

15. Limitations

- Current classification is based on predefined/curated logic.
- It is not a trained machine-learning model.
- Ambiguous descriptions may produce incorrect classification.
- Mixed-material objects can be difficult to classify.
- Waste-management rules differ between locations.
- Image upload currently provides preview functionality only.
- Real-world deployment would require validated AI models and testing.

16. Future Scope

- Real AI/ML waste classification
- Image-based waste recognition
- Multimodal AI
- Voice input
- Multiple languages
- Location-specific disposal rules
- Recycling-center integration
- Mobile application
- User feedback

- Analytics
- Continuous AI model evaluation

17. Conclusion

EcoSort AI demonstrates how a simple digital assistant can help users make better waste-segregation decisions while promoting sustainability awareness. By combining a clear problem statement, alignment with SDG 12 and SDG 11, a working prototype, and responsible AI considerations, the project presents a realistic and honest example of applying technology to a genuine sustainability challenge. While the current prototype relies on curated classification logic rather than a trained AI model, it lays a clear foundation for future enhancement using image recognition, multimodal AI, and location-specific waste-management data.